

Classic Tests of GR

Nonequivalence of Equivalence Principle

close $\rightarrow \frac{1}{r}$

(A) gravitational redshift.

$\Rightarrow$ only uses the equivalence principle

(B) Redshift bending around the sun

$\rightarrow$ total solar eclipse

(C) Precession of Mercury $(V = \frac{Gm}{r} + \frac{Gm^2}{r^2})$

$\rightarrow$ precession

(D) Binary.

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G_{\mu} T_{\mu\nu} = 0$$

$\hookrightarrow$ time independent, spherically symmetric solutions.

$$r = \sqrt{\vec{x} \cdot \vec{x}}$$

$$dx^2 + dy^2 + dz^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2$$

$$ds^2 = A(r) dt^2 + B(r) \vec{x} \cdot d\vec{x} dt +$$

$$C(r) (\vec{x} \cdot d\vec{x})^2 + D(r) (d\vec{x} \cdot d\vec{x})$$

$$\int g_{\mu\nu} dx^\mu dx^\nu$$

$$x^\mu = \begin{pmatrix} t \\ \vec{x} \end{pmatrix} \begin{pmatrix} dt \\ d\vec{x} \end{pmatrix}$$

↓ Redefining Coordinates:

$$T = \frac{1}{2} g^{--} (g_{--}^{-1} g_{--} - g_{--})$$

$$ds^2 = -E(r) dt^2 + F(r) dr^2 + r^2 d\Omega^2$$

$$R_{--} = (\Gamma_{--}^T + \Gamma_{--}^T + \Gamma_{--}^T - \Gamma_{--}^T)$$

$$= -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2$$

$$g^{\mu\nu} (g_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = R - \frac{d}{2} R = -R$$

$$R_{--} = g^{--} R_{--}$$

$$R_{tt} = e^{2(\alpha-\beta)} \left[\frac{d}{dr} \left(\partial^2 \alpha + (\partial \alpha)^2 - \partial \alpha \partial \beta + \frac{2}{r} \partial \alpha \right) \right]$$

$$R_{rr} = -\partial^2 \alpha - (\partial \alpha)^2 + \partial \alpha \partial \beta + \frac{2}{r} \partial \beta$$

$$R_{\theta\theta} = e^{-2\beta} \left[r (\partial \beta - \partial \alpha) - 1 \right] + 1$$

$$R_{\varphi\varphi} = \sin^2 \theta R_{\theta\theta}$$

← Solving for $R_{\mu\nu} = 0$; $\mu = (t, r, \theta, \varphi)$

$$R_{tt} e^{2(\beta-\alpha)} + R_{rr} = \partial^2 \alpha + (\partial \alpha)^2 - \partial \alpha \partial \beta + \frac{2}{r} \partial \alpha - \partial^2 \alpha - (\partial \alpha)^2 + \partial \alpha \partial \beta + \frac{2}{r} \partial \alpha$$

$$= \frac{2}{r} \frac{d}{dr} (\alpha + \beta) \Rightarrow \alpha = -\beta + \text{Constant}$$

$$R_{\theta\theta} = e^{2\alpha} \left[-2r \frac{d\alpha}{dr} - 1 \right] + 1 = 0$$

$$\begin{aligned} \text{Given } \frac{d}{dr}(r e^{2\alpha}) &= e^{2\alpha} + 2r \frac{d\alpha}{dr} e^{2\alpha} \\ &= e^{2\alpha} \left(1 + 2r \frac{d\alpha}{dr} \right) \end{aligned}$$

$$R_{\theta\theta} = -\frac{d}{dr}(r e^{2\alpha}) + 1 = 0$$

$$\frac{d}{dr}(r e^{2\alpha}) = 1$$

$$r e^{2\alpha} = -R_s + r$$

$\Downarrow$

$$e^{2\alpha} = 1 - \frac{R_s}{r}$$

absorb the
constant into
the time
coordinate

$$ds^2 = - \left(1 - \frac{R_s}{r} \right) dt^2 +$$

$$\frac{dr^2}{1 - \frac{R_s}{r}} + r^2 d\Omega^2$$

Weak Field:

$$ds^2 = -(1+2\phi)dt^2 + (1-2\phi)(d\vec{x})^2$$

$$1 + 2\phi \approx 1 - \frac{R_s}{r} \Rightarrow \boxed{R_s = 2G_N M}$$

$\uparrow$
 $-\frac{G_N M}{r}$



$$M = \frac{R_s}{2Gw.}$$

$R_s = 3\text{km}$
for solar mass

$R_s = 0.9\text{cm}$
↑
for Earth mass